

Embedded Language Flows: A Minimalist Bridge Between Image-Domain Flow Matching and Diffusion Language Modeling — A Review of [Hu et al. \[2026\]](#)

Literature review entry

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Abstract

[Hu et al. \[2026\]](#) propose Embedded Language Flows (ELF), a class of continuous diffusion language models (DLMs) based on continuous-time Flow Matching [[Lipman et al., 2023](#), [Liu et al., 2023](#)] operating in a frozen pretrained embedding space. Unlike prior continuous DLMs that supervise discretization at every timestep, ELF keeps the entire sampling trajectory in continuous embedding space and only discretizes at $t = 1$ via the weight-tied unembedding matrix, eliminating the need for a separately trained decoder. The formulation is structurally identical to image-domain latent diffusion [[Rombach et al., 2022](#)], which lets the authors directly transfer classifier-free guidance [[Ho and Salimans, 2022](#)] and other established image-domain techniques. With $\sim 105\text{M}$ parameters, $\sim 10\times$ fewer training tokens than baselines, and fewer sampling steps, ELF outperforms leading discrete DLMs such as MDLM [[Sahoo et al., 2024](#)] and SEDD [[Lou et al., 2024](#)], as well as continuous DLMs descended from Diffusion-LM [[Li et al., 2022](#)]. The result challenges the dominant framing that language requires fundamentally discrete diffusion machinery and offers a cleaner template for cross-modal generative modeling.

Background and Motivation

Diffusion and flow-based generative models [[Ho et al., 2020](#), [Lipman et al., 2023](#), [Liu et al., 2023](#), [Albergo et al., 2025](#)] have become the dominant paradigm for continuous data (images, video, audio). Their extension to language has split into two camps. Discrete DLMs [[Austin et al., 2021](#), [Sahoo et al., 2024](#), [Lou et al., 2024](#)] apply diffusion directly over the categorical token vocabulary, using absorbing-state or score-ratio formulations. Continuous DLMs [[Li et al., 2022](#)] lift discrete tokens into a continuous embedding space, run diffusion there, and discretize back. Continuous DLMs have historically underperformed discrete DLMs because they require (i) per-step cross-entropy supervision to keep the trajectory near valid tokens and (ii) a separately trained decoder that maps embedding-space samples back to text. Both constraints degrade flow flexibility and add architectural overhead.

The paper under review removes both constraints.

Technical Approach

Let $E : \mathcal{V} \rightarrow \mathbb{R}^d$ denote a token embedding map, $z_1 = E(x)$ the clean embedding tensor for sequence $x \in \mathcal{V}^L$, and $z_0 \sim \mathcal{N}(0, I)$. ELF uses a linear interpolation path

$$z_t = (1 - t)z_0 + tz_1, \quad t \in [0, 1]. \quad (1)$$

Flow matching objective. A neural network $v_\theta(z_t, t)$ is trained to predict the target velocity $z_1 - z_0$ via

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{x, z_0, t} \left[\|v_\theta(z_t, t) - (z_1 - z_0)\|^2 \right]. \quad (2)$$

Equivalently the network may predict the clean embedding $\hat{x}_\theta(z_t, t) = \mathbb{E}[z_1 \mid z_t]$ (the x -prediction parameterisation used by the authors).

Final-step discretization via weight-tied unembedding. At inference, z_0 is integrated forward to $t = 1$. ELF’s key observation is that the same encoder that produced $z_1 = E(x) = W_E x$ can be re-used at the final step as the decoder: discrete tokens are recovered via

$$\pi_\theta(z_t \rightarrow 1) = \text{softmax}(W_E \cdot \hat{x}_\theta(z_{t \rightarrow 1}, 1)), \quad (3)$$

where $W_E \in \mathbb{R}^{|\mathcal{V}| \times d}$ is shared between the encoder and the unembedding head. No separate decoder is trained. Cross-entropy supervision is applied *only* at $t = 1$, with weight α , alongside the flow matching loss:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{FM}}(\theta) + \alpha \mathcal{L}_{\text{CE}}(\theta). \quad (4)$$

Classifier-free guidance with self-conditioning. ELF inherits CFG [Ho and Salimans, 2022] directly. The conditioning signal c is the model’s own intermediate prediction (self-conditioning); CFG combines conditional and unconditional velocity fields as

$$v_{\text{cfg}}(z_t \mid c) = \omega v_\theta(z_t \mid c) + (1 - \omega) v_\theta(z_t \mid \emptyset). \quad (5)$$

To avoid the $2\times$ inference cost of naive CFG, the authors adopt *training-time CFG* (cf. image-domain analogues): train the network to output v_{cfg} directly in one forward pass.

Embedding choice. The encoder E is studied in three regimes: pretrained contextual (transformer input embeddings), non-contextual learned (lookup table), and frozen random projection. Pretrained contextual embeddings dominate the ablation, suggesting the well-structured embedding manifold provides much of the lift.

Key Results

ELF is evaluated on three setups. (i) Unconditional generation on OpenWebText, with generative perplexity under a pretrained GPT-2 Large evaluator and unigram entropy as a diversity proxy. (ii) Conditional generation on WMT14 De→En translation. (iii) Conditional generation on summarization.

The headline empirical findings: with a $\sim 105\text{M}$ parameter model trained on $\sim 10\times$ fewer tokens than its $\sim 170\text{M}$ -parameter competitors, ELF achieves lower generative perplexity at fewer sampling steps than MDLM [Sahoo et al., 2024], SEDD [Lou et al., 2024], and prior continuous DLMs. CFG sweeps trace a quality–diversity Pareto front (higher ω lowers perplexity, lowers entropy). Ablations confirm: pretrained contextual embeddings dominate; per-step CE supervision (the prior-art approach) underperforms ELF’s at-final-step-only formulation; training-time CFG matches inference-time CFG quality at half the cost.

Position in the Literature

ELF is a structural simplification of prior continuous DLMs. Where Li et al. [2022] apply per-step categorical supervision to keep the trajectory close to valid tokens, ELF lets the trajectory run free and discretizes only at the final step. Where latent diffusion [Rombach et al., 2022] trains a separate decoder, ELF reuses the encoder via weight tying. Where discrete DLMs [Austin et al., 2021, Sahoo et al., 2024, Lou et al., 2024] avoid the embedding-space question entirely by formulating diffusion over categoricals, ELF returns to the embedding-space formulation and shows it can be made competitive with minimal machinery.

The broader thesis: language can be modeled with the same flow-matching toolkit developed for images, provided the discrete–continuous interface is handled at exactly one boundary (the final time step) rather than diffusely across the trajectory. This is a unification claim in the sense that the empirical machinery is shared with image-domain flow matching [Lipman et al., 2023, Liu et al., 2023].

Limitations and Open Questions

The empirical wins come with caveats. The pretrained contextual embedding is doing significant work; the authors do not quantitatively decompose how much of the gap over discrete baselines is due to the flow matching formulation versus the embedding manifold. Generative perplexity under GPT-2 Large is the headline quality metric, which biases the evaluation toward models whose output distribution resembles GPT-2’s; unigram entropy is a weak diversity proxy that does not catch n -gram-level repetition or mode collapse. No comparison is reported against a similarly-sized autoregressive transformer trained on the same OpenWebText setup with matched token budget — the salient missing control.

The mathematical setup leaves the choice of linear interpolation path unexamined: rectified-flow-style straight-line paths are convenient but not obviously optimal for embedding spaces with non-trivial geometry. The CFG scale ω is swept empirically; no analysis is offered for when the quality–diversity Pareto front saturates or inverts.

Sandbox Probe

A minimal CPU-runnable PyTorch experiment was built to probe the weight-tied-unembedding mechanism in isolation. A tiny MLP is trained to predict $\hat{x}_\theta$ along the linear flow path on a synthetic vocabulary of 64 tokens with 32-dimensional random embeddings; Euler integration with varying step counts is followed by $\arg \max W_E z$ decoding. Source: <https://github.com/pleyva2004/elf-embedded-language-flows> (sandbox subdirectory).

References

- Michael S. Albergo, Nicholas M. Boffi, and Eric Vanden-Eijnden. Stochastic interpolants: A unifying framework for flows and diffusions. *Journal of Machine Learning Research (JMLR)*, 2025.
- Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. Structured denoising diffusion models in discrete state-spaces. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

- Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. In *NeurIPS Workshop on Deep Generative Models and Downstream Applications*, 2022.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- Keya Hu, Linlu Qiu, Yiyang Lu, Hanhong Zhao, Tianhong Li, Yoon Kim, Jacob Andreas, and Kaiming He. ELF: Embedded language flows. *arXiv preprint arXiv:2605.10938*, 2026. URL <https://arxiv.org/abs/2605.10938>.
- Xiang Lisa Li, John Thickstun, Ishaan Gulrajani, Percy Liang, and Tatsunori B. Hashimoto. Diffusion-LM improves controllable text generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023.
- Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *International Conference on Learning Representations (ICLR)*, 2023.
- Aaron Lou, Chenlin Meng, and Stefano Ermon. Discrete diffusion modeling by estimating the ratios of the data distribution. In *International Conference on Machine Learning (ICML)*, 2024.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- Subham Sekhar Sahoo, Marianne Arriola, Yair Schiff, Aaron Gokaslan, Edgar Marroquin, Justin T. Chiu, Alexander Rush, and Volodymyr Kuleshov. Simple and effective masked diffusion language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.